SectraTEK

AGENTIC AI FOR DIGITAL FORENSICS · LAB 01

# Teaching AI to Check Itself



The Reflection Pattern: an AI drafts a forensic report, critiques it, then revises it, so every claim stays inside what the evidence supports.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

**Mobile threat research**
Spyware, stalkerware and device tampering.

**AI for investigators**
Machine learning that speeds up the search.

**Teaching**
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

# Eight steps, one careful habit

1

## What is an agent

A model that follows a loop, not a one-shot answer.

2

## Before AI

How a report was checked by a human reviewer.

3

## The problem

A single AI answer sounds sure, and overclaims.

4

## The discipline

Cite every claim. Separate fact from guess.

5

## The gap

Who reviews the AI when no human is in time?

6

## Reflection

The AI critiques and revises its own draft.

7

## The case

A phone, a leaked file, and six artifacts.

8

## Your lab

Run the loop yourself, free, in the browser.

Practical runs in Google Colab with a free hosted model. Nothing is installed on your computer.

FROM A CHATBOT TO AN INVESTIGATOR

# One answer, or a working loop

## A plain model call

You ask once. The model answers once. If the answer overreaches, nothing catches it.

1. You write a prompt
2. The model replies
3. It stops

TODAY

## An agent

You give the model a goal and a role. It works in a loop: produce, check, and improve before it hands anything back.

1. Produce a draft
2. Check the draft
3. Improve, then repeat

THE MAP OF THE WHOLE COURSE

# Five ways to make a model act like an investigator



## Reflection

Draft, critique, revise.

LAB 01 · TODAY



## Tool Use

Call real tools to inspect evidence.

LAB 02



## ReAct

Reason a step, act, observe, repeat.

LAB 03



## Planning

Break a hard question into steps.

LAB 04



## Multiagent

Split roles, then reconcile findings.

LAB 05

The same idea in five shapes. You are starting with the simplest: the model reviewing itself.

QUALITY CONTROL, THE HUMAN WAY

# How a report used to be checked



## Analyst writes

A first report from the evidence.



## Senior reviews

A second examiner marks overreach and gaps.



## Analyst revises

The report is rewritten to match the evidence.



## Sign-off

A defensible report, hours or days later.

CAREFUL, BUT IT DOES NOT SCALE

This loop works because a human reviewer reads every draft. That is exactly what breaks when a model can write a hundred reports in a minute.

THE PROBLEM

# A confident answer is not a careful one

THE MODEL'S FIRST LINE

**“The employee exfiltrated customers_q1.csv and delivered it to an outside party via Telegram and Gmail.”**

WHY THAT SENTENCE IS A PROBLEM

### No delivery is proven

The artifacts show sending activity, not that anyone received the file.

### No recipient is confirmed

A handle is not a person, and there is no receipt.

### A guess is stated as fact

“Exfiltrated” is a conclusion, written as if it were observed.

THE VOCABULARY THAT KEEPS YOU HONEST

# Observation, inference, claim

## Observation

What an artifact directly records.

**IN THIS CASE**

file_events.csv shows customers_q1.csv created at 01:05:22 UTC.

## Inference

An interpretation that links observations.

**IN THIS CASE**

The file may have been prepared for sharing. Reasonable, but not recorded.

## Claim

A checkable statement placed in the report.

**IN THIS CASE**

“Copied to Telegram at 01:07:41” is supported. “Delivered” is not.

SAY WHAT THE EVIDENCE SAYS

# Cite it, or soften it

| OVERREACHING | | DEFENSIBLE |
|---|---|---|
| “was exfiltrated” | ⇄ | “the artifacts show it was copied and sent” |
| “was delivered to” | ⇄ | “this is consistent with an attempt to share” |
| “the suspect sent it” | ⇄ | “the device recorded an outbound message” |
| “proves theft” | ⇄ | “the current evidence does not confirm receipt” |

The facts are identical. Only the certainty changes, and certainty is what a court tests.

THE GAP

# Who checks the AI, in time?

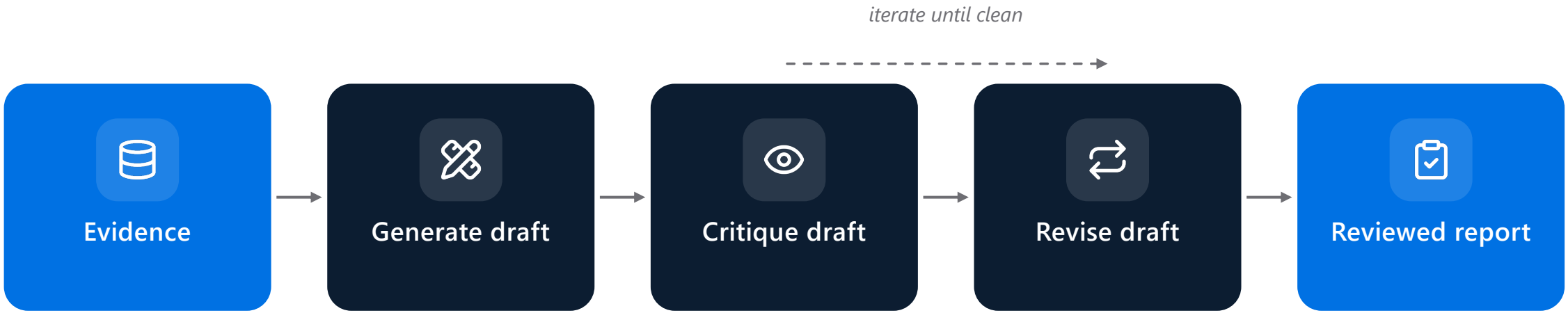



**A MODEL CAN OUTRUN ITS REVIEWERS**

One examiner can read a handful of drafts a day. A model can write hundreds in a minute. Peer review does not scale to that, and a single model call has no built-in check at all. The report ships with whatever overreach it started with.

THE ANSWER

# The Reflection Pattern




**WHEN DOES IT STOP?**

The reviewer returns [[OK]] when nothing material is left to fix, or the loop hits its step limit.

**WHAT YOU GET**

A report whose claims have already survived one honest round of criticism before you ever read it.

THE TRICK BEHIND IT

# One model, three jobs



## The Analyst

Writes the report. Must cite a timestamp and an artifact for every claim.



## The Reviewer

Reads only the draft, not the evidence files. Flags overclaiming and unsourced claims, or says [[OK]].



## The Reviser

The analyst again: it takes the critique and rewrites, with the evidence still in its conversation.

IT IS THE SAME MODEL EACH TIME

Only the instructions change. The role it plays is set by the system prompt. The reviewer works in its own conversation, so it judges the wording of the draft; checking it against the files is your job.

THE INVESTIGATION

# Case DF-2026-017

THE DEVICE

PHONE

Google Pixel 7, Android 14

ACQUIRED

Cellebrite UFED, full file system

WINDOW

overnight, 15 Feb 2026 (UTC)

THE FILE

customers_q1.csv

WHAT THE ARTIFACTS SHOW

## A file appears

customers_q1.csv is created in Downloads and opened.

## It moves and messages fly

The same file is copied into Telegram, then Telegram, Gmail, and network activity follow.

## The original vanishes

customers_q1.csv is deleted from Downloads, all within about seven minutes.

THE EVIDENCE, MINUTE BY MINUTE

# Nine events in about seven minutes

| | |
|---|---|
| 01:05:22 | customers_q1.csv created in Downloads |
| 01:06:12 | same file accessed |
| 01:07:41 | same file hash copied into Telegram storage |
| 01:07:54 | outbound traffic to api.telegram.org |
| 01:07:58 | Telegram message: “sending that sheet now” |
| 01:08:15 | Telegram message: “delete after download” |
| 01:09:39 | outbound SMTP traffic via Gmail |
| 01:09:44 | Gmail message: “see attached export” |
| 01:12:04 | customers_q1.csv deleted from Downloads |

*Shaded rows are the ones people rush to over-read.*

YOUR EVIDENCE PACKAGE

# Six artifacts, one case

### artifact_manifest

Case, device, and integrity metadata.

### file_events

When files were created, copied, deleted, with hashes.

### app_db_messages

Telegram, Gmail, and Slack message excerpts.

### network_events

Outbound connections, bytes, and the app behind them.

### location_events

Approximate device location over time.

### chain_of_custody

Who handled the evidence, and hash verification.

SEE THE LOOP EARN ITS KEEP

# One sentence, three passes

**DRAFT v1**

"The employee exfiltrated the file to an outside party through Telegram and Gmail."

**CRITIQUE**

Over-claimed. The cited events support staging, outbound messaging, and network activity, not confirmed receipt. Separate observed events from inferred transmission.

**REVISED v2**

"Artifacts show the file was copied into Telegram storage, followed by outbound Telegram and Gmail activity consistent with attempted sharing. The evidence does not confirm delivery to a recipient."

KEEP THIS STRAIGHT

# The agent drafts. You decide.



## It speeds the self-check

Many obvious overclaims are caught before a human reads the report.



## It flags, it does not rule

The critique points at risk. It does not establish guilt or innocence.



## It never signs off

A qualified examiner owns the final forensic judgment, always.

THE HONEST FRAMING

Reflection makes a model a better assistant to an examiner. It does not make the model the examiner.

PRACTICAL

# Your turn, in the browser

| | | |
|---|---|---|
| 1 | **Open in Colab** | Click the badge at the top of the notebook. It runs in your browser, nothing to install. |
| 2 | **Part 0: setup** | Steps 0.1 to 0.3: install one add-on, paste a free Groq key (hidden, not saved), write the six case files. |
| 3 | **Part A: by hand** | Steps A.1 to A.5: pack the evidence, get the draft, get the critique, revise. A.6 explains how the loop repeats. |
| 4 | **Part B: the agent** | Steps B.1 to B.3: ReflectionAgent loops draft, review, revise until [[OK]] or 3 rounds, then shows the final report. |
| 5 | **Your Evidence Check** | Four questions: an overclaim in the draft, a claim traced to its artifact and UTC time, the uncertainty wording, what stays human. |

Before every code cell, a short note tells you its Purpose, its Input, How it works, and its Output.

SectraTEK

# Reflection is the first habit of a careful AI.

Next lab: Tool Use, where the agent stops guessing and starts inspecting the artifacts itself.

